SHOAL BATHYMETRY INFLUENCE ON OFFSHORE TRANSPORT AND MIXING

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ABSTRACT

Victor Hieu Le Nguyen: Shoal Bathymetry Influence on Offshore Transport and Mixing
(Under the direction of Dr. Harvey Seim)

Ocean dynamics along coastlines are vastly influenced by topographic features. Particularly, The interactions between shoals and its surrounding flow play an important role in understanding the mixing in the global ocean. Despite this, there remains uncertainty about the role these shoals play in transporting fluid offshore and how they spur vertical and horizontal mixing. In this work, we use a series of large eddy simulations to simulate flow over shoals, analyzing the offshore transport of momentum and tracers, and the generation of eddy kinetic energy upstream and downstream. Our simulations are centered around a reference case of the Diamond Shoals off of North Carolina, using boundary conditions prescribed from observations. Our results show that the shoal serves as an extension of the coastline, diverting northward flow offshore and generating turbulence that enhances mixing. By testing variations in shoal geometry, we find that offshore transport and mixing are driven by the vertical and horizontal extent of the shoal bathymetry relative to the surrounding shelf. Additionally, shoals alter the water column depth and produce a hydraulic jump, transitioning supercritical upstream flow to subcritcal downstream flow.

To my family and friends

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LIST OF ABBREVIATIONS

SAB	South Atlantic Bight
MAB	Middle Atlantic Bight
GS	Gulf Stream
DS	Diamond Shoals
CTD	Conductivity, Temperature, and Depth

1 Introduction

Fluid flow interaction with coastal bathymetry plays a significant role in mass and momentum transfer, which fundamentally drives the movement of nutrients and tracers throughout the ocean [1]. Features such as headlands and capes are also capable of developing eddies that can alter the transport of sediment and dissolved pollutants [2]. Flow-bathymetry interactions therefore have important implications the overall circulation regime of the near-coastal ocean system, and understanding its complex physical dynamics may pave light to new mechanisms relating to transport mechanics in biophysical contexts.

In coastal regions with active tidal currents and rough bathymetry, form drag transfers energy and momentum from large scale to submesoscale. Similarly, tides flowing over bathymetry generate internal waves and eddies that transport mass and removes energy from coastal currents [3]. There is an extensive collection of studies that observe flow interactions with seamounts, headlands, and capes, which are influenced by form drag generated from assymetric bottom pressure across topography [4]. In addition to drag, energy dissipates by wave generation, and wave breaking [5], [6].

The eddies which generate as a result of bathymetry-flow interactions are variable both spatially and temporally. Furthermore, vorticity generation is influenced particularly by torque from stress vertical divergence at bottom bathymetry, and bathymetry induces an intensification and redistribution of vorticity when surrounding waters move towards it [7]. Flow over bathymetry also generate submesoscale structures, which further complicate dynamics in regards to energy and momentum transfer [8].

In the context of flow-bathymetry interactions, shoals are a type of coastal bathymetric feature that have not been studied as extensively as other coastal features such as seamounts, headlands, and capes. Shoals are naturally-generated sandbars and ridges that are submerged in waters located along the coast. Shoals are found on top of continental shelves, and the morphology of shoals could be a potential hazard in maritime transport. Shoals are formed by the gradual accumulation of sand and convergent sediment transport, which can be influenced by both tidal currents and waves [9].

Shoals are susceptible to morphologic change over time, which could be accelerated by changes in sea level, which can further drive changes in coastal circulation and sediment transport.

For this study, we investigate the extent to which shoal bathymetry impacts coastal ocean dynamics as it interacts with flow. Specifically, the impact of shoal bathymetry on offshore transport and mixing can provide new insight to near-coastal circulation dynamics and the role of shoals in mediating the exchange of water between the coast and the open ocean.

The Diamond Shoals extend 13 kilometers out from Cape Hatteras, North Carolina, with a cluster of shallow sandbars that vary in form and depth over time. The Diamond Shoals are infamously known as the ”Graveyard of the Atlantic,” where more than 600 ships have been sunk in this region. The Cape Hatteras coastal region, where the Diamond Shoals lie, are a site of convergent flow between cool, fresh waters from the Middle Atlantic Bight (MAB) north of the shoal, and warm, salty water from the South Atlantic Bight (SAB) south of the Diamond Shoals [10]. Furthermore, the Gulf Stream transitions from a boundary-trapped current to a free jet at The Point, where Cape Hatteras shifts in orientation, forming a dynamic biogeographic boundary with implications on regional carbon transport and biological productivity [11]. Mean circulation and variability taken from observational work on the Cape Hatteras coastal region reveal that the coastal region is influenced by both local wind forcing and variation in the Gulf Stream spatially and temporally [12]

Observational data of waves and currents off Cape Hatteras show that mean subtidal flows are parallel to the coastline on either side of the Cape Hatteras point [13]. Flows in Diamond Shoals follows along the submerged shoal, suggesting that the shoal serves as an extension of the coastline. Along with the Diamond Shoals, the coasts of North Carolina are home to two other shoals, Frying Pan Shoals located off of Cape Fear, and Cape Lookout Shoals located off of Cape Lookout. The presence of these shoals along different parts of the coastline, each with distinct bathymetry and flow pattern, underline the motivation for this study, and we use the Diamond Shoals as a reference case for investigating how shoal bathymetry impacts offshore transport and mixing.

In this paper, we organize as follows. Section 2 introduces background observational data and model setup. Section 3 presents our parameter sweep and detailed analysis on offshore flux, eddy kinetic energy, and baroclinicity. In Section 4, we comment on overall shoal-fluid dynamics and present our results in broader context.

2 Methods

We use the Diamond Shoals as a reference case to analyze shoal morphology on ocean dynamics. While focusing on specific parameters and setup similar to the Diamond Shoals, it should be noted that this study is generalizable and our model can be tuned to observe offshore dynamics and energetics on a variety of shoal-shelf morphology, which will be further discussed in Section 4.

2.1 Observational Data

Observational data for this study were derived from the Processes driving Exchange At Cape Hatteras program, which used moorings, gliders, ADCP, and HF radar over an 18-month period to observe exchange between shelf-water and open ocean in Cape Hatteras, North Carolina [10]. For this study, we use processed CTD data from two moorings located roughly 50 km north (B1) and south (B2) of the Point as depicted in Fig. 1.

The B1 mooring is located at $35.77820°N$, $75.09470°W$, situated north of The Point. Mean temperature and salinity for this mooring at 5 m depth are roughly $20.5°C$ and 32.6 psu, respectively. In contrast, the B2 mooring is located at $34.78230°N$, $75.94120°W$ southwest of the The Point, with mean temperature and salinity at 5 m depth being roughly $24.5°C$ and 35.5 psu, respectively (Fig. 2). Here, southward transport over the MAB shelf transports relatively cold and fresh water southward toward Cape Hatteras. In contrast to the B1 mooring, the B2 mooring is located where the SAB shelf transports relatively warm, salty water northward towards The Point [14]. Mean temperature and salinity profiles from B1 and B2 suggest a stronger density gradient at the surface than at depth (Fig. 3).

We use GEBCO_2014 gridded bathymetry data [15] to determine the shelf-shoal morphology of the Diamond Shoals, and construct a profile of the shoal using a piecewise function with sigmoidal smoothing. By taking a 40 km transect starting from the point of Cape Hatteras towards offshore,

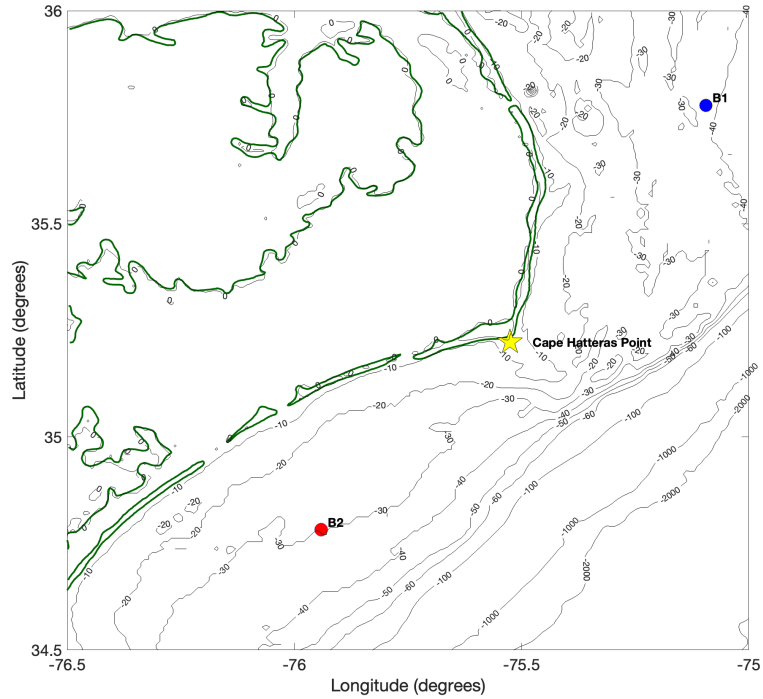

Figure 1: Map of B1 (north) and B2 (south) moorings deployed during PEACH project from 2017-2018. Green lines represent the coastline, and the point of Cape Hatteras is indicated as a yellow star. Bathymetry data used to form the contour lines are from GEBCO.

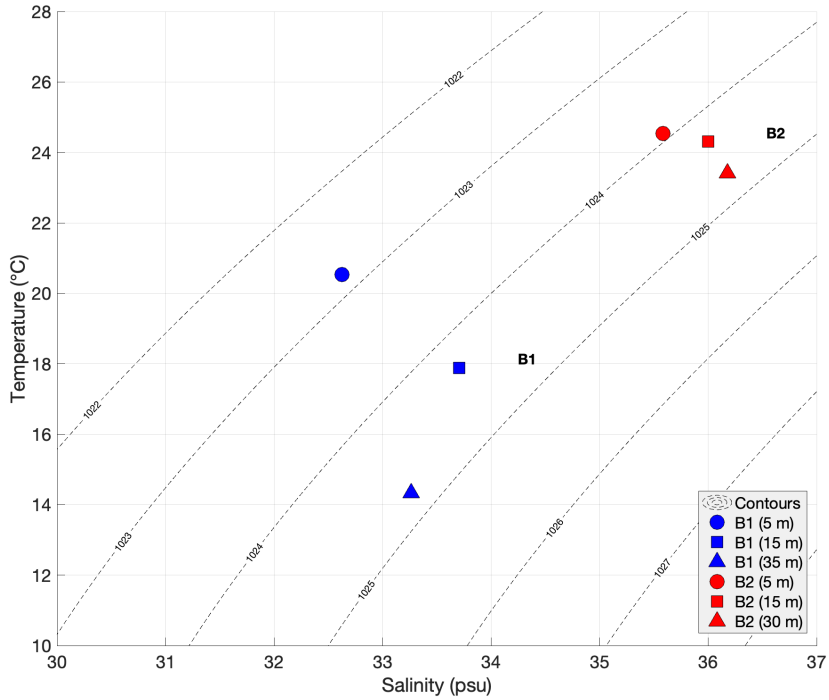

Figure 2: B1 and B2 Temperature/Salinity for 5 m (circle), 15 m (square), 30/35 m (triangle) depths. Density contour lines are overlaid.

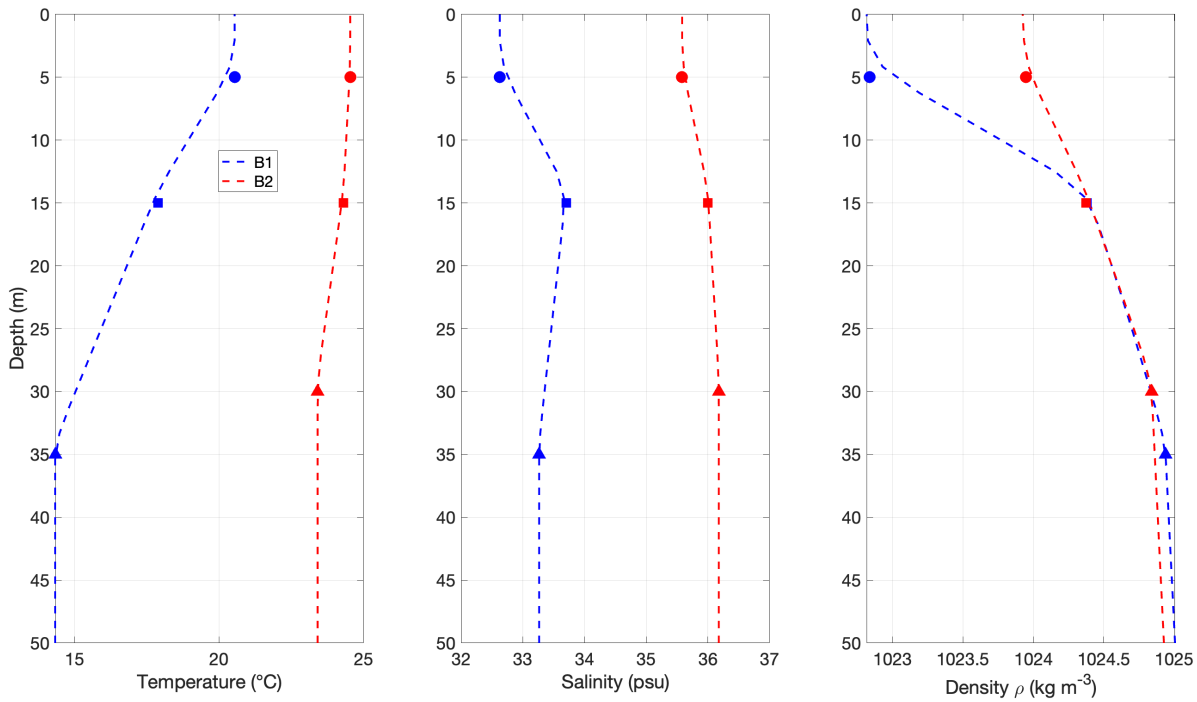

Figure 3: Interpolated temperature, salinity, and density profiles for B1 (blue) and B2 (red) moorings deployed during PEACH. Marked symbols indicate the surface (circle), middle (square), and bottom (triangle) depths in which CTD data was collected. Density is calculated using TEOS-10.

the shoal itself can be broken down into 4 main slopes: 1) a steep coastal ramp connected directly to the coastline and extending to 5 m depth, 2) the top of the shoal spanning 20 km offshore from 5 m to 15 m depth, 3) a steep shelf slope from 15 m depth to 50 m depth, and 4) a flat surface. It should be noted that we limit depth from 0 to 50 m to isolate shoal-shelf dynamics rather than deeper offshore ones (Fig. 4).

2.2 Model Setup

Our idealized flow over shoals model was developed using Oceananigans, a Julia package which solves the nonhydrostatic incompressible Boussinesq equations for fluid flow [16]. Oceananigans uses finite-volume discretizations to solve the Boussinesq equations, where density of the modeled fluid is decomposed into a constant reference density ρ_0 and a perturbation density ρ',

$$\rho(x, y, z, t) = \rho_0 + \rho'(x, y, z, t) \tag{1}$$

With this approximation, the momentum equations are written as

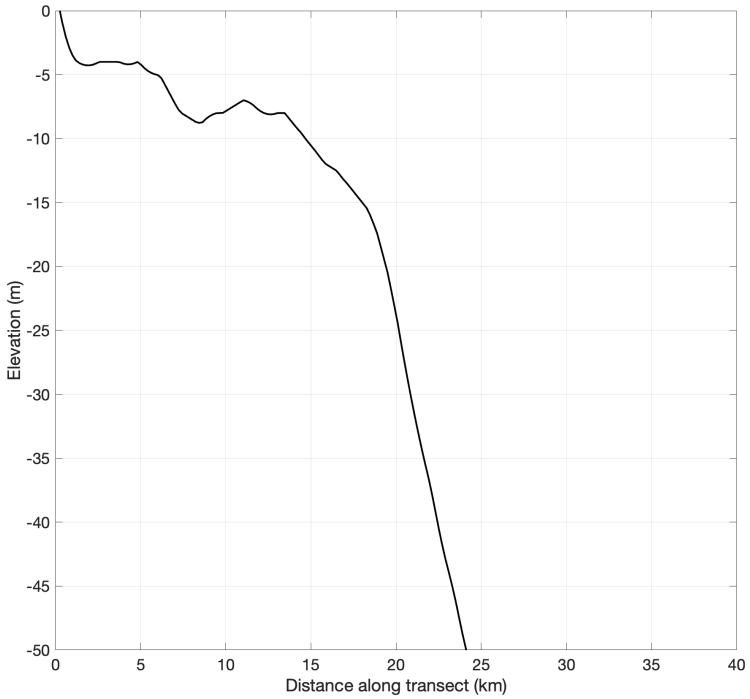

Figure 4: X-Z (along-shoal) transect of Diamond Shoals with GEBCO bathymetry data.

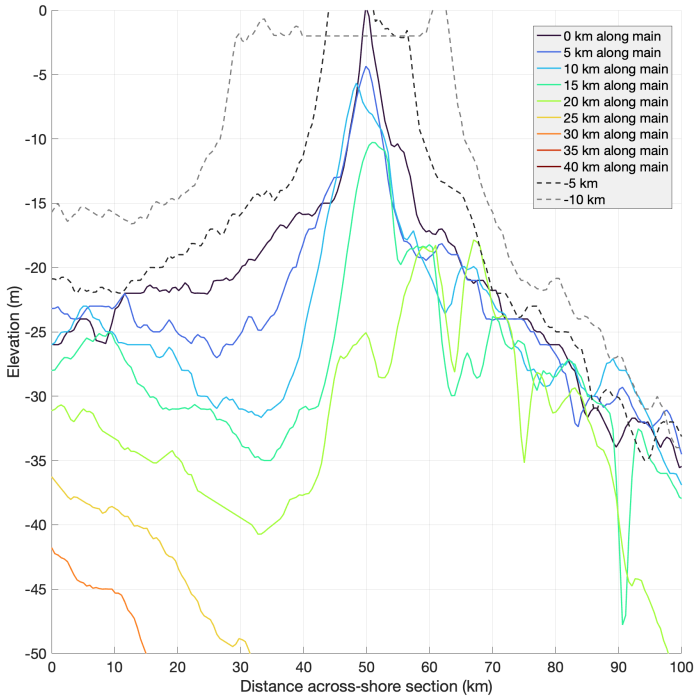

Figure 5: Y-Z (across-shoal) transects of Diamond Shoals with GEBCO bathymetry data, taken at 5 km intervals along the shoal. The shoal is centered at on the x axis at 50 km across shore.

$$\partial_t \boldsymbol{u} = -\boldsymbol{\nabla} p - \underbrace{(\boldsymbol{u}\cdot\boldsymbol{\nabla})\boldsymbol{u} - (\boldsymbol{u}_g\cdot\boldsymbol{\nabla})\boldsymbol{u} - (\boldsymbol{u}\cdot\boldsymbol{\nabla})\boldsymbol{u}_g}_{\text{advection}} - \underbrace{\boldsymbol{f}\times\boldsymbol{u}}_{\text{Coriolis}}$$
$$+ \underbrace{(\boldsymbol{\nabla}\times\boldsymbol{u}_s)\times\boldsymbol{u} + \partial_t\boldsymbol{u}_s}_{\text{Stokes drift}} - \underbrace{b\hat{\boldsymbol{g}}}_{\text{buoyancy}} - \underbrace{\boldsymbol{\nabla}\cdot\boldsymbol{\tau}}_{\text{closure}} + \underbrace{\boldsymbol{F}_u}_{\text{forcing}}, \tag{2}$$

where $\boldsymbol{u} = (u, v, w)$ is the Eulerian velocity, p is the kinematic pressure, $\boldsymbol{u}_g$ is a background geostrophic flow, $\boldsymbol{f} = (0, 0, f)$ is the Coriolis parameter, $\boldsymbol{u}_s$ is the Stokes drift, $b = -g\rho'/\rho_0$ is the buoyancy, $\boldsymbol{\tau}$ is the turbulent stress tensor, and $\boldsymbol{F}_u$ represents additional forcing.

In all of our simulations, we use a 5th-order Weighted Essentially Non-Oscillatory (WENO) advection scheme and a 3rd-order Runge-Kutta time-stepping method. We also use an anisotropic minimum dissipation (AMD) model as a turbulence closure and a conjugate gradient method to solve for the pressure on the immersed bathymetry boundary. The choice of advection scheme, timestepping method, turbulence closure, and pressure solver are based off the headland flow model setup from [4], also written in Oceananigans. For all simulations, the model is run for 100 days.

Initial bathymetry setup was developed using topographic data from General Bathymetric Chart of the Oceans (GEBCO). An XZ slice of the Diamond Shoals was taken, and extrapolated into a piecewise, linear curve, to represent the cross-shore bathymetry, and combined with a gaussian bump, an idealized bathymetry of the Diamond Shoals was fitted into the model grid with a horizontal resolution (Nx, Ny) of 500 m, and a vertical resolution (Nz) of 1 m. It should be noted that the orientation of Cape Hatteras is $10°N$ on the east side, $166°N$ at the cape point, and $120°N$ farther west. For simplicity, the shelf and shoal is placed along the western wall, so along-shoal direction refers to position along the x-axis, and cross-shoal direction refers to position along the y-axis.

As a reference case, idealized model bathymetry is constructed with shoal height $H_S = 15$ m, protruding upward in the z dimension relative to the start of the shelf, and shoal length $L_S = 20$ km, extending from the beginning of the shelf offshore in the x direction (Fig. 6). Shoal width σ_S, which measures the distance along-shore from the center of the shoal to its extent in the y direction, is 8

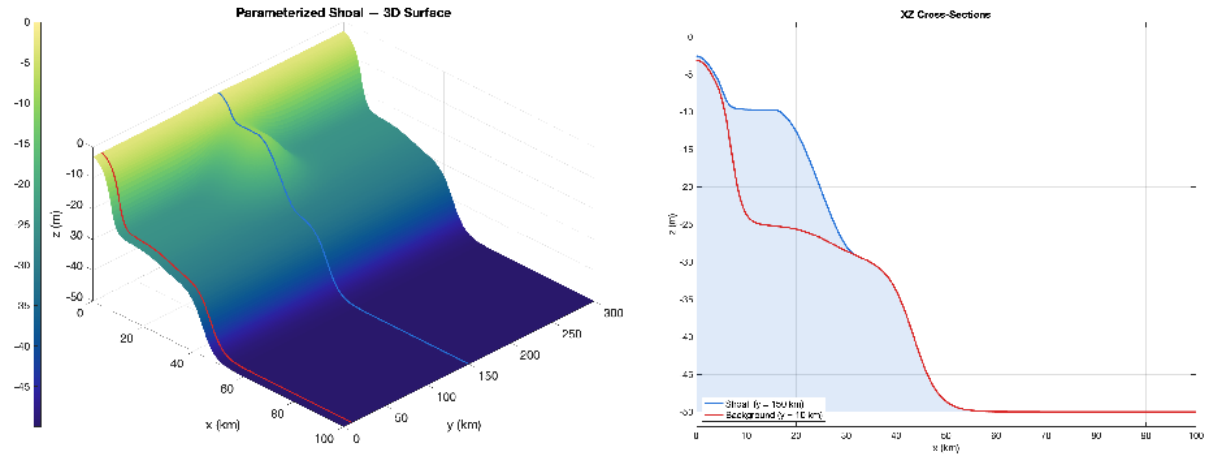

Figure 6: (Left) Idealized 3D bathymetry of the Diamond Shoals. (Right) X-Z transect of the background continental shelf (red) and the idealized shoal (blue).

km for our reference case. Shelf width L_{sh} measures the distance offshore that the shelf protrudes, and is set at 50 km. Lastly, shelf height H_{sh} is set at 25 m from the bottom of the domain.

For boundary conditions, we use periodic boundary conditions in the north and south (y-direction), so flow moving towards the north will return through the southern boundary as it passes through. In the reference case, we prescribe a background flow of $v_0 = 0.1$ m/s in the y direction, representing a mean northward flow across the shoal. In the case of one directional flow, temperature and salinity boundary conditions are value (dirichlet) boundary conditions matching the temperature and salinity profile from the B2 mooring south of Cape Hatteras. To better constrain offshore transport, we nudge the offshore region of the domain (where $x > L_{sh}$) towards zero velocity and a stratified temperature and salinity profile. For this reference case, wind forcing is not considered. For all simulations, the Coriolis force is implemented as an f-plane approximation, where $f = 8.4168 \times 10^{-5}$ s^{-1}.

3 Results

To diagnose the impact of shoal bathymetry on offshore transport and mixing, we define our output variables below.

Offshore flow can be represented by velocity in the x-direction (offshore), which we call u. For a positive u, currents are moving offshore, and the opposite for a negative u. In our simulation

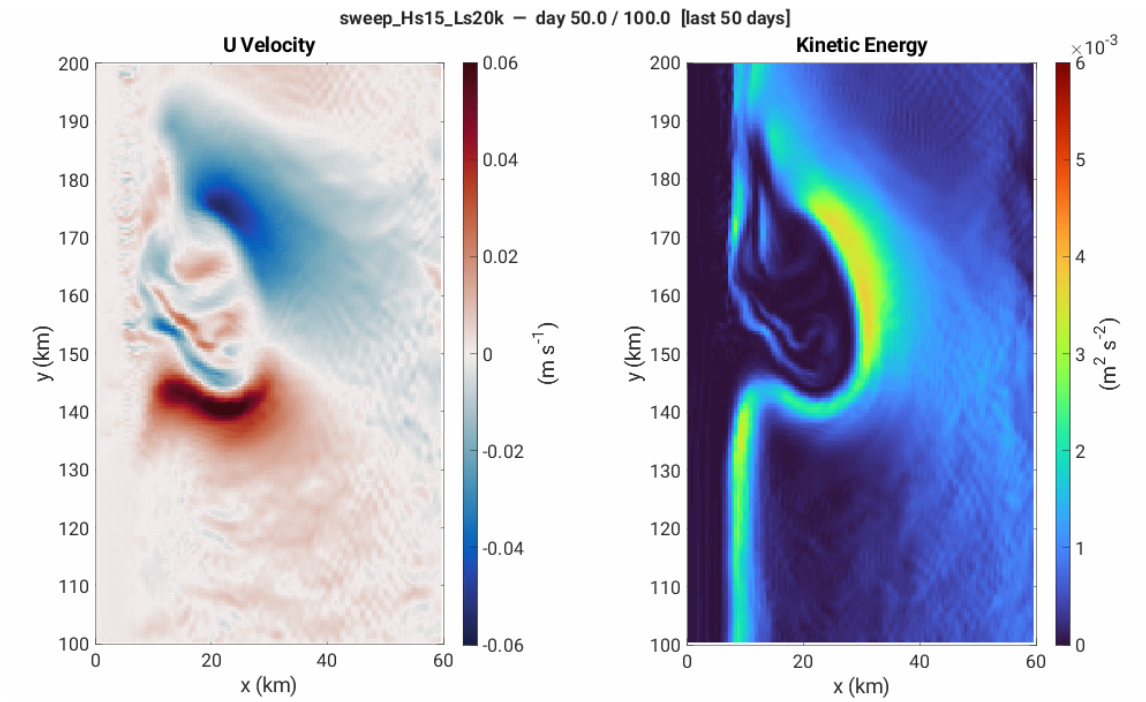

Figure 7: Surface (z = 0) u-velocity (left) and kinetic energy (right) for the reference case. X axis represents distance offshore (km), and y axis represents distance alongshore (km).

figures, offshore flow represents flow moving towards the east (right), and onshore flow represents flow moving towards the west (left).

In addition to offshore flow shown in our simulation output (Fig. 7), we also calculate kinetic energy from the velocity field, which is defined as

$$KE = \frac{1}{2}(u^2 + v^2 + w^2) \tag{3}$$

where u, v, w are the velocity components in the x, y, and z directions, respectively.

To quantify the transport of heat and salt, we calculate the time-averaged offshore flux. The time-averaged flux of a tracer ϕ (representing temperature T or salinity S) is defined as:

$$\overline{u\phi} = \frac{1}{\Delta t}\int_{t_0}^{t_0+\Delta t} u(x,y,z,t)\phi(x,y,z,t)dt \tag{4}$$

where we use an averaging period of $\Delta t = 10$ days. Furthermore, we compute the depth-averaged offshore flux as:

$$\langle \overline{u\phi} \rangle_z = \frac{1}{H} \int_{-H}^{0} \overline{u\phi} dz \tag{5}$$

where H is the local water depth.

Lastly, we analyze the turbulent kinetic energy (TKE) to investigate flow instabilities and mixing near the shoal. Using Reynolds decomposition ($u = \bar{u} + u'$), the time-averaged TKE is defined as:

$$\overline{TKE} = \frac{1}{2} \left[(\overline{u^2} - \bar{u}^2) + (\overline{v^2} - \bar{v}^2) + (\overline{w^2} - \bar{w}^2) \right] \tag{6}$$

where the overline denotes a time-average.

For our reference case, as background flow moving northward encounters the shoal, it is deflected offshore upstream. In the region where the shoal is present on the shelf, the diverted flow follows along the shoal, and the velocity near the shoal is higher than the background velocity along the rest of the shelf. Comparing the upstream (toward the shoal) and downstream (past the shoal) velocity, offshore velocity concentrates upstream at the edge of the shoal, while onshore velocity is spread out further in the y direction downstream of the shoal. The assymetry between these two patterns suggests that the turbulence generated on the shoal as flow moves toward it redirects flow offshore. High kinetic energy is present on the surface in regions along the linear coastal ramp, and along the edge of the shoal in the x direction. Notably, the appearance of eddies on the surface, over the top of the shoal, suggests turbulent flow over the shoal that can mix the water column vertically and horizontally.

3.1 Parameter Sweep

To better understand how shoal bathymetry alters flow dynamics, we conduct a parameter sweep of bathymetry-related variables. For the scope of this study, we analyze the effect of shoal height H_S and shoal length L_S. Constants and our changing parameters are listed in Table 1.

3.2 Offshore Flux

TODO:

Table 1: Simulation parameters for the parameter sweep. Bracketed parameters represent different test cases, with the reference case bolded.

Symbol	**Parameter**	**Value(s)**
L_x	Domain length (x)	100 km
L_y	Domain width (y)	300 km
L_z	Domain depth (z)	50 m
f	Coriolis frequency (f-plane)	8.4168×10^{-5} s^{-1}
H_S	Shoal height	[5, 10, **15**] m
L_S	Shoal length (x-direction)	[**20**, 30, 40] km
σ_S	Shoal width (y-direction)	8 km
L_{sh}	Shelf width (x-direction)	50 km
H_{sh}	Shelf depth	25 m
(V_N, V_S)	North–South inflow velocity	(0.1, 0.1) m s^{-1}

calculate volume transport? transport at shelf break? Plots for this:

last time averaged window of offshore velocity (x-direction) on xy map for varying shoal heights and varying shoal lengths.

time averaged window of offshore transport ($uT - uT_{east}$ and $uS - uS_{east}$) ? Try using recent cluster set from a couple days ago.

stacked line plot of depth averaged and y averaged (over clipped window) offshore velocity (u) for varying shoal heights and varying shoal lengths.

Should get a sense of shallower shoals -¿ more offshore transport? Longer shoal -¿ larger volume transport? Larger barrier for mean northward flow.

avg volume transport as a dot plot where x axis is shoal length and y axis is volume transport.

similar idea for shaol height? maybe some covariance.

3.3 Eddy Kinetic Energy

TODO:

mean kinetic energy and eddy/turbulent kinetic energy depth averaged and time averaged plots.

maybe look at mid-depth slices for KE?

show clear transition of MKE to EKE/TKE as side by side panels.

4 Discussion

4.1 Future Work

4.2 Implications

5 Conclusion

general summary of question, method, findings

5.1 Future Work

additional parameter sweep (bathymetry and external forcing), model resolution.

5.2 Implications

open the door for further shoal-related modeling studies using Oceananigans. GPU computing allows for higher resolution and more complex physics. Nutrient transport? BGC?

1 An Appendix

2 A Second Appendix

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